

## Space Complexity

Sunday, 22 March 2026 8:51 AM

### Measure of space taken by your code.

Write a code to return array containing square of n numbers

```
int[] SQ(int n)
{
    int[] ans = new int[n];
    int[] ret = new int[n];
    int i;
    for(i=1; i<=n; i++)
    {
        ret[i-1] = i * i;
    }
    return ret;
}
```

Array of size n  $\rightarrow$  n unit space  
1 unit

$$TC = n + 1 \approx O(n)$$

TC comes into picture :

1. When we define data structure of size n
2. When we have recursion  $\rightarrow$  Because of recursive stack trace.

$$\boxed{m \mid n \leq}$$

int [][] arr = new int[m][n]

Dictionary / Hashmap - Not available in C

int [][] arr = new int[m][n]  $\rightarrow$   $SC = O(m \times n)$

$\uparrow$   
m rows  
n columns

$$\left. \begin{array}{l} T(2^n) \\ T(n^2) \end{array} \right\}$$

For FAANGs user experience is most important  $\Rightarrow$  Lower TC is preferred  
We do tradeoff of time with space  $\Rightarrow$  We use extra memory to improve TC

$\Rightarrow$  in DP we use extra array/dictionary to reduce TC from  $O(2^n) \rightarrow O(n^2)$

